12 kW Wind Turbine M8

Build book · Ø8.0 m rotor · 8 m tower · revision A

Item	M8 on the 8 m tower
Rated power	12 kW electric at 10.0 m/s (hub height)
Rotor	Ø8.0 m, 3 glass/epoxy blades with carbon caps, TSR 5.5, 131 rpm, collective pitch
Regulation	variable speed + one electric pitch cylinder (supercap backed); feather in storms
Generator	Rev D direct-drive PM: Rev C + 2 magnet rows, stack 104 mm, 28 turns
Brake	3 spring-applied calipers, 14.4 kNm, rotor lock pin
Hub height	8.0 m; lowest blade tip 4.0 m above ground
Tower	one-piece tilt-up monopole Ø400 → Ø400, t 10/8, 671 kg, f1 2.78 Hz
Foundation	spread footing 3.0–3.4 m square, 6.2–7.8 m³; rock plinth 2 m³
Head mass	1362 kg above the tower flange (CAD, every part)
Energy (vs 18 m tower)	sheltered 13.4 MWh (−42 %) · inland 24.4 MWh (−27 %) · coastal 35.9 MWh (−18 %) · highland 47.8 MWh (−16 %)



Contents: design summary (§1); general arrangement and BOM (§2); part catalogue with an axonometric view of every part (§3); the NM8 drawing set rev A (§4); reused drawings (§5); the assembly sequence in 13 steps (§6); erection, commissioning and inspection (§7); open items (§8).

Sources: docs 35 and 37, analysis/variant_m8 (dlc8_m8.py, tower8_m8.py, spindle_m8.py, hub_yaw_m8.py, generator_m8.py), cad/turbine_m8 (model8.py, drawings8.py, iso8.py, bom8.py), cad/hub_m8. Every number comes from those scripts; re-run them to update the book.

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How to use this book

- Every part has a card in §3 (view, part number, material, mass, drawing). The drawing follows in §4 (new and changed M8 parts) or §5 (reused unchanged).
- Build in the order of §6. Each step lists the parts, the torques (NM8-900) and the checks that must be signed before the next step.
- Site work (footing, erection, commissioning) is §7. Keep the signed checklists with the turbine file.
- All drawings are NOT CHECKED. Release for the prototype only after the review and the checks of §8.

1 Design summary: 12 kW M8 on an 8 m tower

What the 8 m tower changes

The M8 turbine of doc 35 is unchanged above the yaw bearing except for what the detail design of doc 37 found: the third brake caliper moved to 135° (the rotor lock holds 90°), the hub rear gussets were shortened to clear the drum web, the spinner grew to Ø1 040 to enclose the pitch bearings, and the nacelle cover was lengthened 15 mm at the front.

Tower. A soft-stiff tower is impossible at 8 m, so the tower is stiff: f1 above 1.15 × 1P at 131 rpm and below 3P at rated. The 3P excitation crosses f1 at low speed, so the controller passes that speed band under torque.

Tower check	Value
Shell	Ø400 → Ø400, t 10/8, 671 kg
f1 rigid / soft soil + ice	2.78 / 2.65 Hz (window 2.51–5.95)
3P exclusion zone	48–61 rpm
Governing case	earthquake 0.5 g (ground D) + operation, utilisation 0.768
Fatigue damage (base)	0.53
Base bolts	16 × M24 10.9 on Ø500
Tilt-up	150.0 kNm, line pull 66.2 kN (gin pole 0.33 H)

Loads. The whole aeroelastic load set was rerun at 8 m hub height with stronger shear and turbulence and the stiff tower. Simulated at 8 m (103 cases, shear 0.25, turbulence +20 %, stiff tower): Mtilt 16.0, thrust 6.8 (design values, doc 37 §2). The 18 m controller tripped often in the stronger turbulence and once reached 220 rpm, so the 8 m controller has a faster speed loop (6 rad/s), a faster normal pitch rate (20°/s) and the trip at 165 rpm; peak 181 rpm (design 184). Blade, spindle, bearings, brake and tower were re-checked with the larger of the 18 m and 8 m envelopes; the brake grows to 14.4 kNm.

Energy at 8 m

Slower, more turbulent wind near the ground. Log wind profile per site roughness:

Site (roughness z0)	Mean wind 18 m → 8 m	AEP 18 m	AEP 8 m	Loss	Turbulence
Sheltered valley / farm (0.25 m)	5.0 → 4.05 m/s	23.0 MWh	13.4 MWh	−42 %	+23 %
Open inland lowland (0.05 m)	6.0 → 5.17 m/s	33.4 MWh	24.4 MWh	−27 %	+16 %
Coast / exposed lowland (0.01 m)	7.0 → 6.24 m/s	43.8 MWh	35.9 MWh	−18 %	+12 %
Highland / headland (0.03 m)	8.5 → 7.42 m/s	57.0 MWh	47.8 MWh	−16 %	+14 %

- Coast, highland and open land lose 16–27 %; a sheltered farm loses 42 % and should use the 18 m tower.
- Safety: lowest blade tip 4.0 m above ground. Fenced keep-out radius 5 m; ice-fall zone 24 m when icing is possible.

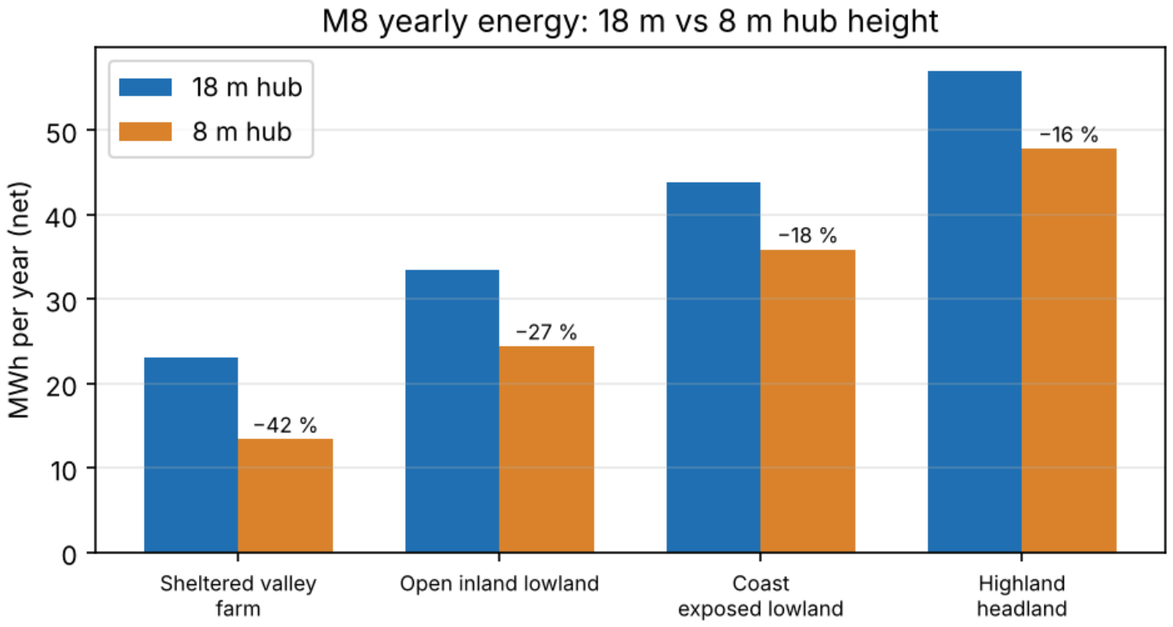
Decisions used in this book (doc 35 defaults, open for Martin)

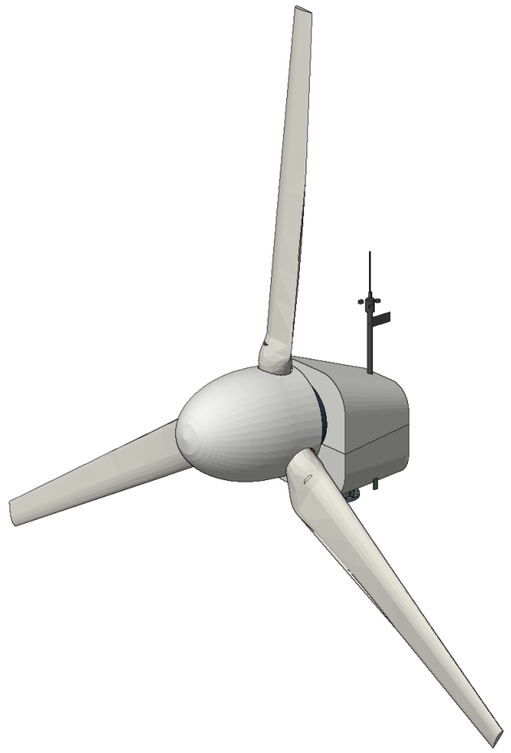
- D1 bedplate NP8-300 (136 kg) instead of NP1-300 (338 kg).
- D2 three calipers 4.8 kNm (third at 135°) instead of two stronger ones.
- D3 plain carbon caps; D4 keep 28 turns; D5 crank r140 (pin plane moved to 150, clash-free).

Main parts

- Blades 3 × 34.1 kg; hub and pitch 337 kg budget; generator Rev D; NP8-300 bedplate; yaw ring Dpw 410.
- Spindle reserve 2.5, bearing s0 6.7/6.2, life factor 5.6.

1 Design summary: figures and masses





Mass by group (assembled turbine)

Group	kg (CAD)
nacelle	170
electrical	134
yaw	288
drivetrain	440
rotor	455
tower	741
foundation	15042
Head above the tower flange	1362

CAD head mass includes every modelled part (cover, frame, adapter, lock holder, junction box, spinner); the doc 35 budget was 1,280 kg. The tower was checked with the larger value.